

Technology #160

Add a micro■actuated counterweight system and
a closed■loop control algorithm that adjusts
its position based on the imbalance sensor
output, integrated into the spindle drive

Engineering Technology-Transfer Dossier
electronics

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-10

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Add a micro-actuated counterweight system and a closed-loop control algorithm that adjusts its position based on the imbalance sensor output, integrated into the spindle drive electronics

WHY DOES IT MATTER?

Documented problem 'rotor imbalance' concerning benchtop centrifuge module. User-submitted engineering problem (session input): "design a compact benchtop centrifuge module for sealed micro-vials with an aerodynamic rotor enclosure, a brushless spindle drive, an imbalance sensor, and a lid-interlock safety circuit". ROUTING: doe_osti: routing state SUSPENDED_RELEVANCE — measured relevance instability (R375: same fusion papers relevant 3 -> 0; pooled relevant-rate 0.175 pre-fix; DETERMINISM_GAP flagged in the R394 production audit) — records poison synthesis-feeding retrieval more than they feed it until relevance is semantically adjudicated; reinstatement: the Phase P1 semantic reranker lands (semantic relevance adjudication between query and record, replacing lexical overlap); flip to ACTIVE + a measured relevance re-run + worklog entry. [fra_rail_accidents] retrieval status GRAMMAR_MISMATCH (query does not match source grammar: YYYY-MM-DD date floor (\$where date >= X) — engine routing error, not a provider failure (not absence, Art. XXV)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] retrieval status NOT_QUERIED_SUSPENDED_RELEVANCE (engine routing decision (R399 W3): SUSPENDED_RELEVANCE — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 23 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in ml_data. 5-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

constraint-compliance envelope article

WHAT WOULD MAKE US KILL IT?

killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 5-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: constraint-compliance envelope article

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - constraint-compliance envelope article - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: constraint-compliance characterization. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-002 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: held-out + temporal validation splits. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-004 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: failure-mode stress tests (distribution shift). Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data; empirical risk on a non-representative sample says nothing about deployment risk (distribution shift); utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured; ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.; ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.; ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.; Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate $[y_i, -]$ value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: constraint-compliance envelope article. Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Real-time active mass redistribution using a miniature movable counterweight driven by a micro■actuator, guided by the imbalance sensor to counteract rotor asymmetry

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'rotor imbalance' concerning benchtop centrifuge module. User-submitted engineering problem (session input): "design a compact benchtop centrifuge module for sealed micro-vials with an aerodynamic rotor enclosure, a brushless spindle drive, an imbalance sensor, and a lid-interlock safety circuit". ROUTING: doe_osti: routing state SUSPENDED_RELEVANCE — measured relevance instability (R375: same fusion papers relevant 3 -> 0; pooled relevant-rate 0.175 pre-fix; DETERMINISM_GAP flagged in the R394 production audit) — records poison synthesis-feeding retrieval more than they feed it until relevance is semantically adjudicated; reinstatement: the Phase P1 semantic reranker lands (semantic relevance adjudication between query and record, replacing lexical overlap); flip to ACTIVE + a measured relevance re-run + worklog entry. [fra_rail_accidents] retrieval status GRAMMAR_MISMATCH (query does not match source grammar: YYYY-MM-DD date floor (\$where date >= X) — engine routing error, not a provider failure (not absence, Art. XXV)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] retrieval status NOT_QUERIED_SUSPENDED_RELEVANCE (engine routing decision (R399 W3): SUSPENDED_RELEVANCE — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: ml_data

Disciplines: data engineering, machine learning, software engineering, statistics

Architecture: PROPOSED architecture (no physical system exists): data ingestion and labeling pipeline; feature / representation layer; model + uncertainty quantification; deployment target + monitoring loop — realizing: Add a micro-actuated counterweight system and a closed-loop control algorithm that adjusts its position based on the imbalance sensor output, integrated into the spindle drive electronics

Subsystems:

?: data ingestion and labeling pipeline - ?

?: feature / representation layer - ?

?: model + uncertainty quantification - ?

?: deployment target + monitoring loop - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Add a micro-actuated counterweight system and a closed-loop control algorithm that adjusts its position based on the imbalance sensor output, integrated into the spindle drive electronics

Key Physics: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

ML-002: Bayes' theorem (posterior inference). Expression: $P(y|x) = P(x|y) * P(y) / P(x)$. Applicability to this invention: CONDITIONAL; holds under: probability model correctly specified; priors explicit. Assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data. Engaged variables: NOT ESTABLISHED.

ML-003: Empirical risk. Expression: $R_{emp} = \text{sum}(\text{loss}(f(x_i), y_i)) / n$. Applicability to this invention: APPLICABLE; holds under: samples independent and representative of the deployment distribution. Assumptions: empirical risk on a non-representative sample says nothing about deployment risk (distribution shift). Engaged variables: $n, f(x_i)$.

ML-001: Expected utility of a decision rule. Expression: $EU = \text{sum}_y P(y | x) * U(a(x), y)$. Applicability to this invention: CONDITIONAL; holds under: utility table fully specified; probabilities calibrated. Assumptions: utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured. Engaged variables: NOT ESTABLISHED.

Assumptions:

- priors and likelihoods are MODEL_DERIVED until validated on deployment data
- empirical risk on a non-representative sample says nothing about deployment risk (distribution shift)
- utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured
- ML-002 (Bayes' theorem (posterior inference)) is applicable only under: probability model correctly specified; priors explicit.
- ML-003 (Empirical risk) is applicable only under: samples independent and representative of the deployment distribution.
- ML-001 (Expected utility of a decision rule) is applicable only under: utility table fully specified; probabilities calibrated.
- Model input variables requiring sourced values before any numeric result: $P(x|y)$ (-), $P(y)$ (-), $P(x)$ (-), n (-), loss (-), $f(x_i)$ (-), y_i (-), $P(y|x)$ (-), U (-) — all UNKNOWN (no sourced value exists; Art. VIII).

- Model output variables produced by the governing model: $P(y|x)$ (-), R_{emp} (-), EU (-) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): training data volume and provenance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; label quality rate $[y_i, -]$ value_status=UNKNOWN; target operating point (precision/recall) [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; calibration error budget [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; inference latency ceiling [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; distribution-shift monitoring thresholds [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'rotor imbalance', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'distribution shift degrading accuracy post deployment', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'label noise propagating to biased predictions', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'overfitting to source-site artifacts', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'calibration drift invalidating decision thresholds', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'feedback-loop data contamination', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	compact design	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Real-time active mass redistribution using a miniature movable counterweight	MODELED
DI-003	Expected effect target	Measurable reduction in vibration amplitude and rotor speed ripple, enabling	MODELED
DI-004	Domain design input pattern: decision	in the applicability of support tokens [‘decision’]	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: available	applicability of ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: latent	variability of ESTABLISHED (candidate pattern; confirm again)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance	with candidates for the 21st Century Cures Act (21st Century Cures Act) is never inferred	ENGINEERING_PROPOSED
DI-008	Custodied observation: Repurposing	of a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-009	Custodied observation: Development	of a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-010	Custodied observation: A low-cost	bandwidth controller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-011	Custodied observation: Assessing	the impact of a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-012	Custodied observation: Decitabine	cytotoxicity for a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-013	Custodied observation: A multi-wavelength	laser for a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-014	Custodied observation: Analysis	of a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-015	Custodied observation: Centrifuge	for a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-016	Custodied observation: A Wi-Fi	streaming controller for a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-017	Custodied observation: A continuous	streaming controller for a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-018	Custodied observation: Development	of a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-019	Custodied observation: A Protocol	for a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-020	Custodied observation: The development	of a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-021	Custodied observation: Dependence	of a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-022	Custodied observation: Rotor imbalance	suppression for a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED
DI-023	Custodied observation: Dynamics	analysis for a single-core microcontroller for a single-core microcontroller	ENGINEERING_PROPOSED

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	data ingestion and labeling pipeline realizing Add a micro	actuated counterweight	system architecture
DO-002	feature / representation layer realizing Add a micro	actuated counterweight	system architecture
DO-003	model + uncertainty quantification realizing Add a micro	actuated counterweight	system architecture
DO-004	deployment target + monitoring loop realizing Add a micro	actuated counterweight	system architecture
DO-005	data ingestion and labeling pipeline interfaces with feature	representation layer	signal processing (name of the data stream)
DO-006	feature / representation layer interfaces with model + uncertainty	quantification	signal processing (name of the data stream)
DO-007	model + uncertainty quantification interfaces with deployment	target + monitoring loop	signal processing (name of the data stream)
DO-008	Device boundaries: patient/fluid inlet and outlet terminal	control/data processing	signal processing (name of the data stream)
DO-009	Flow/force/signal path geometry realizing the mechanism	of a single-core micro	actuated counterweight

ID	Description	Status	Missing Inputs
DO-010	Operating range for training data volume and provenance	UNKNOWN	Which the mechanism holds its stated behavior for training data volume and provenance;
DO-011	Operating range for label quality rate within which the mechanism holds its stated behavior	UNKNOWN	Which the mechanism holds its stated behavior for label quality rate; verification of range e
DO-012	Operating range for target operating point (precision/recall)	UNKNOWN	Which the mechanism holds its stated behavior for target operating point (precision/recall)
DO-013	Operating range for calibration error budget within which the mechanism holds its stated behavior	UNKNOWN	Which the mechanism holds its stated behavior for calibration error budget; verification of
DO-014	Operating range for inference latency ceiling within which the mechanism holds its stated behavior	UNKNOWN	Which the mechanism holds its stated behavior for inference latency ceiling; verification of
DO-015	Operating range for distribution-shift monitoring thresholds	UNKNOWN	Which the mechanism holds its stated behavior for distribution-shift monitoring thresholds;
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism's operating point	PROPOSED	Input/output, sensing modality, actuated, decision thresholds (source values), proposed
DO-017	Information pipeline: ingestion -> feature/representation	PROPOSED	UQ -> decision stage, input/output, sensing loop, latency budget, times, training, test, edge spec
DO-018	Fixture enabling: held-out + temporal validation splits	UNKNOWN	Not named, build specification, UNKNOWN selection; acceptance instrumentation
DO-019	Fixture enabling: calibration and decision-threshold analysis	UNKNOWN	Not named, build specification, UNKNOWN selection; acceptance instrumentation
DO-020	Fixture enabling: failure-mode stress tests (distribution shift)	UNKNOWN	Not named, build specification, UNKNOWN selection; acceptance instrumentation
DO-021	Fixture enabling: held-out + temporal split evaluation	UNKNOWN	Not named, build specification, UNKNOWN selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
training data volume and distribution	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as source candidate) value and verification method L
label quality rate	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design as source candidate) value and verification method L
target operating point (pressure)	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as source candidate) value and verification method L
calibration error budget	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as source candidate) value and verification method L
inference latency ceiling	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as source candidate) value and verification method L
distribution-shift monitoring thresholds	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design as source candidate) value and verification method L

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
rotor imbalance	Fails by rotor imbalance: the design failure mechanism is a failure of the rotor, designed with a new operating tested, considered failure problem (Armed)	design failure mechanism is a failure of the rotor, designed with a new operating tested, considered failure problem (Armed)	design failure mechanism is a failure of the rotor, designed with a new operating tested, considered failure problem (Armed)	design failure mechanism is a failure of the rotor, designed with a new operating tested, considered failure problem (Armed)
distribution shift degradation	Fails by distribution shift degradation and yet to be deployed, the deployment shift stress testing QW test passed, candidate fatigue is mode (Armed)	Fails by distribution shift degradation and yet to be deployed, the deployment shift stress testing QW test passed, candidate fatigue is mode (Armed)	Fails by distribution shift degradation and yet to be deployed, the deployment shift stress testing QW test passed, candidate fatigue is mode (Armed)	Fails by distribution shift degradation and yet to be deployed, the deployment shift stress testing QW test passed, candidate fatigue is mode (Armed)
label noise propagating	Fails by label noise propagating to biased predictions, the label noise propagation is a failure of the model, considered failure problem (Armed)	Fails by label noise propagating to biased predictions, the label noise propagation is a failure of the model, considered failure problem (Armed)	Fails by label noise propagating to biased predictions, the label noise propagation is a failure of the model, considered failure problem (Armed)	Fails by label noise propagating to biased predictions, the label noise propagation is a failure of the model, considered failure problem (Armed)
overfitting to source	Fails by overfitting to source, the training data is a failure of the model, considered failure problem (Armed)	Fails by overfitting to source, the training data is a failure of the model, considered failure problem (Armed)	Fails by overfitting to source, the training data is a failure of the model, considered failure problem (Armed)	Fails by overfitting to source, the training data is a failure of the model, considered failure problem (Armed)
calibration drift invalid	Fails by calibration drift invalid, the decision threshold is a failure of the model, considered failure problem (Armed)	Fails by calibration drift invalid, the decision threshold is a failure of the model, considered failure problem (Armed)	Fails by calibration drift invalid, the decision threshold is a failure of the model, considered failure problem (Armed)	Fails by calibration drift invalid, the decision threshold is a failure of the model, considered failure problem (Armed)
feedback-loop data corruption	Fails by feedback-loop data external adjudication sampling, the data is a failure of the model, considered failure problem (Armed)	Fails by feedback-loop data external adjudication sampling, the data is a failure of the model, considered failure problem (Armed)	Fails by feedback-loop data external adjudication sampling, the data is a failure of the model, considered failure problem (Armed)	Fails by feedback-loop data external adjudication sampling, the data is a failure of the model, considered failure problem (Armed)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
rotor imbalance	Fails by rotor imbalance: the device fails to maintain balanced rotation	Design controls for the rotor assembly; regular maintenance and balancing	Residual imbalance due to manufacturing tolerances and wear
distribution shift degrading accuracy	Fails by distribution shift: the model's performance degrades on new data	Robust training techniques; domain adaptation; active learning for feedback	Unseen data distributions; adversarial examples
label noise propagating to predictions	Fails by label noise: incorrect labels in training data propagate to model	Noise detection and correction; robust loss functions; synthetic data augmentation	Unavoidable noise in real-world data collection
overfitting to source-site artifacts	Fails by overfitting to source-site artifacts, not generalizing to target sites	Regularization; cross-site validation; data augmentation; transfer learning	Site-specific artifacts; distributional differences
calibration drift invalidating decisions	Fails by calibration drift: model's confidence becomes unreliable over time	Continuous monitoring; recalibration; ensemble methods; uncertainty quantification	Drift in model parameters; changing data characteristics
feedback-loop data contamination	Fails by feedback-loop data contamination: model's output influences its input	External validation; holdout sets; synthetic data; adversarial testing	Model's own output; feedback loop; data poisoning

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Run the centrifuge with a deliberately introduced 5% g-force imbalance deliberately introduced 5% g-force imbalance	Run the centrifuge with a deliberately introduced 5% g-force imbalance	Run the centrifuge with a deliberately introduced 5% g-force imbalance	Run the centrifuge with a deliberately introduced 5% g-force imbalance
VF-002	held-out + temporal validation splits. Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%	held-out + temporal validation splits. Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%	held-out + temporal validation splits. Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%	held-out + temporal validation splits. Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%
VF-003	held-out + temporal validation splits. Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%	held-out + temporal validation splits. Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%	held-out + temporal validation splits. Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%	held-out + temporal validation splits. Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%
VF-004	failure-mode stress tests (distribution shift). Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%	failure-mode stress tests (distribution shift). Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%	failure-mode stress tests (distribution shift). Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%	failure-mode stress tests (distribution shift). Acceptance criterion: (ENGINEERING_PROTOCOLS) - 100% (ENGINEERING_PROTOCOLS) - 100%

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification pass)	criteria to be fixed at NOT PERFORMED	NOT PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
NOT_APPLICABLE — data only invention: no physical materials exist to select; hardware dependencies are recorded under design	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
NOT	ESTABLISHED (no domain data)	NOT	ESTABLISHED (no domain data)				

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- MLOps CI for data and model: domain registry manufacturing route - candidate only; no process qualification exists
- versioned artifact registry with rollback: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-004', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter training data volume and provenance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter label quality rate [y_i -]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter target operating point (precision/recall) [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter calibration error budget [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter inference latency ceiling [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter distribution-shift monitoring thresholds [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}

- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	constraint-compliance envelope per article method	per method	constraint-compliance envelope per article method	NO TESTS ESTABLISHED (to be quoted)	constraint-compliance envelope per article method	NO TESTS ESTABLISHED (to be quoted)
WP-02	article for VF-001	per method	Run the centrifuge with article for VF-001	NO TESTS ESTABLISHED (to be quoted)	VF-001 verification article for VF-001	NO TESTS ESTABLISHED (to be quoted)
WP-03	article for VF-002	per method	held-out + temporal variation per article method	NO TESTS ESTABLISHED (to be quoted)	VF-002 verification article for VF-002	NO TESTS ESTABLISHED (to be quoted)
WP-04	article for VF-003	per method	held-out + temporal variation per article method	NO TESTS ESTABLISHED (to be quoted)	VF-003 verification article for VF-003	NO TESTS ESTABLISHED (to be quoted)
WP-05	article for VF-004	per method	failure-mode stress test per article method	NO TESTS ESTABLISHED (to be quoted)	VF-004 verification article for VF-004	NO TESTS ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 5-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.